

Isabella Mann

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Computer Science researcher and developer with specialized experience at NASA and Oregon State University labs. Adept at cross-disciplinary collaboration, conducting real-world experimental trials, and engineering tools that accelerate scientific data visualization. Seeking a technical role within the public sector or national laboratories to support mission-critical research.

Education

Oregon State University — Bachelor of Science in Computer Science (Data Science Focus)

Expected Graduation: June 2026 | GPA: 3.05

Experience

Software Engineer Intern | NASA Langley Research Center, Hampton, VA

June 2023–August 2023

- Engineered novel data visualization tools within the Atmospheric Science Data Center to interpret complex datasets from the TEMPO satellite.
- Migrated legacy atmospheric data to ArcGIS, enabling real-time web rendering.
- Co-engineered a Python-based web mapping service integrating Tempo satellite data, reducing data visualization time for researchers and users.

Research Software Developer | Sustainable, Healthy, and Resilient Buildings Lab, Corvallis, OR February 2024–Present

- Engineered a centralized, autonomous data-ingestion pipeline by repurposing legacy laboratory hardware into a remote-controlled API server, automating real-time telemetry aggregation for a network of 50+ sensors.
- Developed and integrated automated data-cleaning protocols using R, transforming high-dimensional raw sensor outputs into analysis ready datasets, reducing latency between data collection and scientific analysis.
- Lead the design of complex, multi-faceted visual analytics that translate high-volume metrics into accessible, statistically valid graphics, bridging the gap between raw data and scientific insights for diverse stakeholders.
- Collaborated with interdisciplinary research teams to define analytical requirements for indoor air quality, aerosol dynamics, and wildfire smoke research, accelerating cross disciplinary hypothesis generation.

Software Testing Engineer Intern | Trimble, Lake Oswego, OR

June 2025–September 2025

- Engineered multiple RESTful API endpoints to facilitate efficient data retrieval and operations with MongoDB.
- Developed 200+ automated API and UI tests, improving test coverage and supporting product functionality.
- Created and merged over 17 pull requests in GitHub, contributing directly to the main product codebase.
- Collaborated with other interns and engineers to design and develop efficient, scalable testing solutions using Agile software development methodologies and processes.

Undergraduate Researcher | Oregon State University, Corvallis, OR

February 2023–June 2023

- Worked in a small team to further research on the OptiBand, a wearable assistive device for visually impaired users, by analyzing datasets, conducting real-world experiments, and preparing academic publications.
- During the program's timeline, conducted real-world studies in which individuals with vision disabilities used the OptiBand in their day-to-day lives and answered surveys about their experiences.

Projects

Senior Capstone | Year Long Academic Project

September 2025–June 2026

- Collaborated in a multidisciplinary team of Computer Science and Electrical Engineering to create an open-source device that translates sound into tactile vibrations for the deaf and hard of hearing community.
- Engineered a full-stack WiFi web app to provide a user-friendly interface for an audio-haptic wearable device.
- Built the front-end UI/UX using React/Preact, creating an intuitive dashboard focusing on human-centered design for users to configure haptic response behaviors and audio analysis settings.

Applied Deep Learning | Winter 2026 Academic Project

January 2026–March 2026

- Engineered a ConvNeXt model block-by-block for accurate image classification on the CIFAR-10 dataset and implemented robust experiment logging.
- Developed generative computer vision models, including a Variational Autoencoder (VAE) for image synthesis.
- Designed sequence-to-sequence models leveraging Recurrent Neural Networks (RNNs) and Attention mechanisms for complex sequence processing tasks.
- Implemented a Transformer-based language model to replicate modern Large Language Model architectures.

Skills

- Languages:** C, C++, CSS, HTML5, Kotlin, JavaScript, R, Python, SQL, TypeScript | Intermediate Spanish
- Frameworks & Libraries:** React, Preact, PyTorch, Pandas, NumPy, Node.js
- Tools & Environments:** Linux, Git/GitHub, Visual Studio Code, PyCharm, PowerShell, Playwright, Postman, Swagger API, MongoDB, Jira, Slack, Microsoft SharePoint & Teams, HPC Environments

